

separation function

$$\tau_{B \times_f F}((p, x), (q, y)) := \sup L(\gamma)$$

where the sup is w.r.t. all future directed causal curves γ that connect (p, x) and (q, y) . If there is no such curve, one sets $\tau((p, x), (q, y)) = 0$. $\tau_{B \times_f F}$ satisfies the reverse Δ -inequality

$$\tau_{B \times_f F}((p, x), (q, y)) \geq \tau_{B \times_f F}((p, x), (r, z)) + \tau_{B \times_f F}((r, z), (q, y)).$$

For all $(p, x), (q, y), (r, z) \in I \times F$. In the following we write also $\tau_{B \times_f F} = \tau$.

The time separation function has the following properties:

- $\tau(y, y') = 0$ if y' and y are not causally related,
- $\tau(y, y') > 0$ if $y \ll y'$.

The generalized Lorentzian cone w.r.t. $-I$, the metric space F and the function $f : I \rightarrow (0, \infty)$ is

$$(I \times F, D, \ll, \leq, \tau) =: -I \times_f F.$$

If F is an intrinsic metric space, then $-I \times_f F$ is a Lorentzian pre-length space in the sense of [KS18].

Here we collect some properties of $-I \times_f F$. We always assume that F is intrinsic.

- $-I \times_f F$ has the push-up property: if $(s, x) \ll (t, y)$ if and only if there exists a future directed causal curve from (s, x) to (t, y) with positive length. Moreover $I^{+/-}((s, x))$ is open for all $(s, t) \in I \times F$.
- If F is geodesic, then $J^\pm((s, x))$ is closed for all $(s, x) \in I \times F$.

The next theorem is the Lorentzian analogue of Theorem 2.5 above.

Theorem 3.1 (Kunzinger, Saemann, Graf, Alexander). *Let X be a strictly intrinsic metric space and let $\gamma = (\alpha, \beta)$ be a future directed, causal curve that is a maximizer w.r.t. L in $-I \times_f F$ and parametrized proportional to arclength. Then*

- (a) β is a minimizer in F ;
- (b) (Fiber independence) α is independent of F , except for the total height, i.e. the length $L^F(\beta)$ of β . More precisely, if $\hat{F}$ is another strictly intrinsic metric space and $\hat{\beta}$ is a minimizing geodesic in $\bar{F}$ with the same length and speed as β , then $(\alpha, \hat{\beta})$ is a minimizer in $-I \times_f \hat{F}$.
- (c) If γ is timelike, then β has speed $\frac{c_\gamma}{2f^2\alpha}$ for a constant c_γ .
- (d) If γ is timelike, then $-\frac{1}{2}(\alpha')^2 + \frac{1}{2f^2\alpha} = E$ for a constant E .

We stated above that a generalized cone $-I \times_f F$ over an intrinsic metric space F is a Lorentzian pre-length space. If F is locally compact, it was shown in [AGKS23] that $-I \times_f F$ is even a Lorentzian length spaces.

Theorem 3.2. *Let F be a locally compact length space. Then $-I \times_f F$ is a strongly causal Lorentzian length space.*